

2025 UU IM Code and RRT Analysis

```
#install.packages('here')
library(readxl)
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(here)
```

here() starts at /Users/reblocke/Research/U of U IM Codes and Rapid Responses/Publications/ATS 2026 Abstract

```
#folder <- "C:/Code Projects/Code Blue Training Year 2/Data/"
#file <- "arcbt.xlsx"
#survey_responses <- read_excel(paste0(folder, file))

# changed this to a relative path
survey_responses <- readxl::read_excel(here::here("data", "arcbt.xlsx"))
```

New names:

- `` -> `...1`

```
survey_responses <- survey_responses %>%
  mutate(
    change_con_code = post_con_code - pre_con_code,
    change_con_rapid = post_con_rapid - pre_con_rapid,
    change_con_rhythm = post_con_rhythm - pre_con_rhythm,
    change_attending = post_attending - pre_attending,
    change_team = post_team - pre_team
  )

summary(survey_responses)
```

...1	pre_con_code	pre_con_rapid	pre_con_rhythm	pre_team
Length:22	Min. :1.000	Min. :2.000	Min. :2.000	Min. :2
Class :character	1st Qu.:3.000	1st Qu.:4.000	1st Qu.:4.000	1st Qu.:4
Mode :character	Median :5.000	Median :5.000	Median :6.000	Median :5

	Mean :4.318	Mean :5.227	Mean :5.545	Mean :5
	3rd Qu.:5.000	3rd Qu.:6.750	3rd Qu.:7.000	3rd Qu.:6
	Max. :7.000	Max. :7.000	Max. :8.000	Max. :9
pre_attending	pre_codes	pre_rapid	pre_lead	
Min. :1.000	Min. :0.000	Min. : 7.00	Min. :0.000	
1st Qu.:3.000	1st Qu.:2.000	1st Qu.:10.00	1st Qu.:1.000	
Median :3.000	Median :3.000	Median :15.00	Median :2.000	
Mean :4.045	Mean :3.045	Mean :15.50	Mean :2.636	
3rd Qu.:5.000	3rd Qu.:4.000	3rd Qu.:18.75	3rd Qu.:4.750	
Max. :9.000	Max. :7.000	Max. :30.00	Max. :8.000	
pre_sim	pre_shock	post_con_code	post_con_rapid	
Min. :0.000	Length:22	Min. :4.000	Min. :5.000	
1st Qu.:0.250	Class :character	1st Qu.:5.250	1st Qu.:6.000	
Median :1.000	Mode :character	Median :7.000	Median :7.000	
Mean :1.409		Mean :6.682	Mean :6.909	
3rd Qu.:2.000		3rd Qu.:8.000	3rd Qu.:8.000	
Max. :5.000		Max. :8.000	Max. :9.000	
post_con_rhythm	post_team	post_attending	post_modules	
Min. :5.000	Min. :2.000	Min. : 1.000	Min. : 6.000	
1st Qu.:6.250	1st Qu.:3.000	1st Qu.: 3.000	1st Qu.: 8.000	
Median :7.000	Median :4.000	Median : 3.500	Median : 8.500	
Mean :7.182	Mean :4.591	Mean : 4.091	Mean : 8.545	
3rd Qu.:8.000	3rd Qu.:5.000	3rd Qu.: 5.000	3rd Qu.: 9.750	
Max. :9.000	Max. :9.000	Max. :10.000	Max. :10.000	
post_cases	post_discuss	change_con_code	change_con_rapid	
Min. : 7.000	Min. : 4.000	Min. : -1.000	Min. :0.000	
1st Qu.: 8.000	1st Qu.: 7.250	1st Qu.: 1.250	1st Qu.:1.000	
Median : 8.500	Median : 8.500	Median : 2.000	Median :2.000	
Mean : 8.636	Mean : 8.455	Mean : 2.364	Mean :1.682	
3rd Qu.: 9.750	3rd Qu.:10.000	3rd Qu.: 3.000	3rd Qu.:2.000	
Max. :10.000	Max. :10.000	Max. : 6.000	Max. :4.000	
change_con_rhythm	change_attending	change_team		
Min. : -1.000	Min. : -3.000000	Min. : -5.00000		
1st Qu.: 0.250	1st Qu.: -1.000000	1st Qu.: -1.7500		
Median : 2.000	Median : 0.000000	Median : -1.0000		
Mean : 1.636	Mean : 0.04545	Mean : -0.4091		
3rd Qu.: 3.000	3rd Qu.: 1.000000	3rd Qu.: 0.7500		
Max. : 4.000	Max. : 3.000000	Max. : 6.0000		

```
library(tinytable)
tt(survey_responses)
```

...1	pre_con_code	pre_con_rapid	pre_con_rhthm	pre_team	pre_attendi
B4019	7	7	7	5	3
J3331	3	5	8	4	5
G0675	5	7	7	9	4

...1	pre_con_code	pre_con_rapid	pre_con_rhthm	pre_team	pre_attendi
A1184	4	4	4	3	3
G8087	6	4	4	5	9
D0695	5	6	6	5	3
S8893	5	5	4	3	3
G8879	5	6	8	5	3
M5075	6	6	7	5	3
C5096	2	3	6	6	1
B4221	2	7	2	5	4
S2807	5	7	6	6	2
L9099	4	5	5	8	3
S5625	5	7	7	4	6
Pitt8657	3	5	3	7	3
D4268	3	4	4	6	7
M6411	5	5	8	2	3
H1619	6	7	6	5	6
R9991	2	3	4	5	4
Q2614	6	4	4	6	5
B5084	5	6	6	4	2
S3369	1	2	6	2	7

Pre

```
library(modelsummary)

# Define custom functions for the 25th and 75th percentiles
q1 <- function(x) quantile(x, 0.25, na.rm = TRUE)
q3 <- function(x) quantile(x, 0.75, na.rm = TRUE)
n_non_missing <- function(x) sum(!is.na(x))

# Create the summary table with median, Q1, Q3, and N
datasummary((N = n_non_missing) +
```

```
(Median) +
(Q1 = q1) +
(Q3 = q3) ~
  pre_con_code + pre_con_rapid + pre_con_rhthm + pre_team + pre_attending + p
data = survey_responses)
```

	pre_con_code	pre_con_rapid	pre_con_rhthm	pre_team	pre_attendin
N	22.00	22.00	22.00	22.00	22.0
Median	5.00	5.00	6.00	5.00	3.0
Q1	3.00	4.00	4.00	4.00	3.0
Q3	5.00	6.75	7.00	6.00	5.0

Post

```
datasummary((N = n_non_missing) +
  (Median) +
  (Q1 = q1) +
  (Q3 = q3) ~
    post_con_code + post_con_rapid + post_con_rhythm + post_team + post_attendi
data = survey_responses)
```

	post_con_code	post_con_rapid	post_con_rhythm	post_team	post_a
N	22.00	22.00	22.00	22.00	
Median	7.00	7.00	7.00	4.00	
Q1	5.25	6.00	6.25	3.00	
Q3	8.00	8.00	8.00	5.00	

Change in Scores

```
datasummary((N = n_non_missing) +
  (Median = median) +
  (Q1 = q1) +
  (Q3 = q3) ~
    change_con_code + change_con_rapid + change_con_rhythm + change_attending +
data = survey_responses)
```

	change_con_code	change_con_rapid	change_con_rhythm	change_att
N	22.00	22.00	22.00	
Median	2.00	2.00	2.00	
Q1	1.25	1.00	0.25	
Q3	3.00	2.00	3.00	

```
datasummary_skim(survey_responses)
```

	Unique	Missing Pct.	Mean	SD	Min	Median	Max	Histogram
pre_con_code	7	0	4.3	1.6	1.0	5.0	7.0	
pre_con_rapid	6	0	5.2	1.5	2.0	5.0	7.0	
pre_con_rhthm	7	0	5.5	1.7	2.0	6.0	8.0	
pre_team	8	0	5.0	1.7	2.0	5.0	9.0	
pre_attending	8	0	4.0	1.9	1.0	3.0	9.0	
pre_codes	8	0	3.0	1.6	0.0	3.0	7.0	
pre_rapid	8	0	15.5	7.0	7.0	15.0	30.0	
pre_lead	7	0	2.6	2.2	0.0	2.0	8.0	
pre_sim	5	0	1.4	1.3	0.0	1.0	5.0	
post_con_code	5	0	6.7	1.3	4.0	7.0	8.0	
post_con_rapid	5	0	6.9	1.3	5.0	7.0	9.0	
post_con_rhythm	5	0	7.2	1.1	5.0	7.0	9.0	
post_team	7	0	4.6	2.1	2.0	4.0	9.0	
post_attending	8	0	4.1	2.2	1.0	3.5	10.0	
post_modules	5	0	8.5	1.2	6.0	8.5	10.0	
post_cases	4	0	8.6	1.0	7.0	8.5	10.0	
post_discuss	5	0	8.5	1.6	4.0	8.5	10.0	

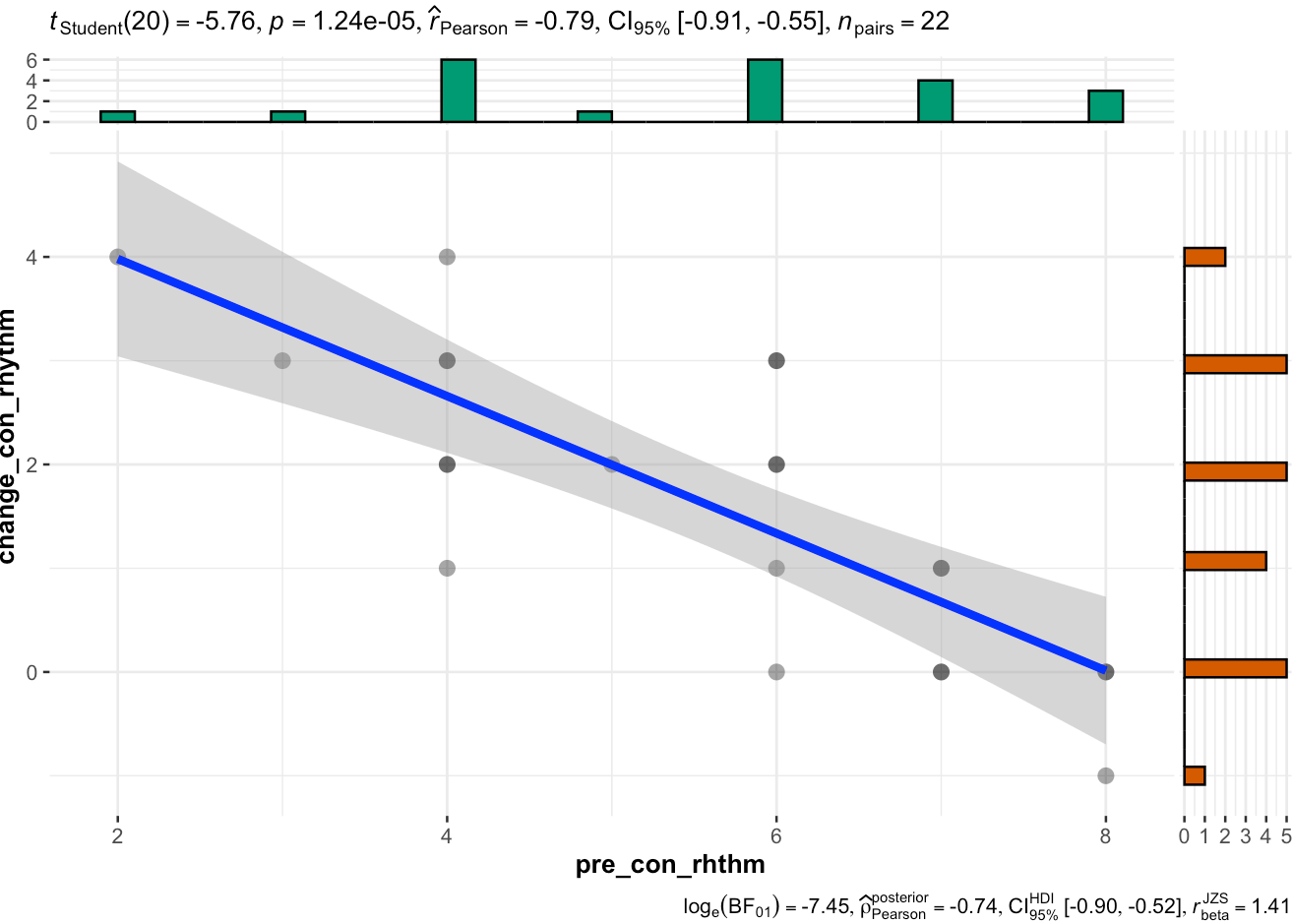
	Unique	Missing Pct.	Mean	SD	Min	Median	Max	Histogra
change_con_code	6	0	2.4	1.6	-1.0	2.0	6.0	
change_con_rapid	5	0	1.7	1.0	0.0	2.0	4.0	
change_con_rhythm	6	0	1.6	1.4	-1.0	2.0	4.0	
change_attending	7	0	0.0	1.4	-3.0	0.0	3.0	
change_team	9	0	-0.4	2.5	-5.0	-1.0	6.0	

		N	%
...1	A1184	1	4.5
	B4019	1	4.5
	B4221	1	4.5
	B5084	1	4.5
	C5096	1	4.5
	D0695	1	4.5
	D4268	1	4.5
	G0675	1	4.5
	G8087	1	4.5
	G8879	1	4.5
	H1619	1	4.5
	J3331	1	4.5
	L9099	1	4.5
	M5075	1	4.5
	M6411	1	4.5
	Pitt8657	1	4.5
	Q2614	1	4.5
	R9991	1	4.5
	S2807	1	4.5

	Unique	Missing	Mean	SD	Min	Median	Max	Histogram
		Pct.						
pre_shock	S3369	1	4.5					
	S5625	1	4.5					
	S8893	1	4.5					
	No	16	72.7					
	Yes	6	27.3					

```
ggstatsplot::ggscatterstats(data = survey_responses, x = pre_con_rhthm, y = change_con_rh
```

`stat\_xsidebin()` using `bins = 30`. Pick better value `binwidth`.
`stat\_ysidebin()` using `bins = 30`. Pick better value `binwidth`.



```
t.test(survey_responses$post_con_rhythm, survey_responses$pre_con_rhthm, paired = TRUE)
```

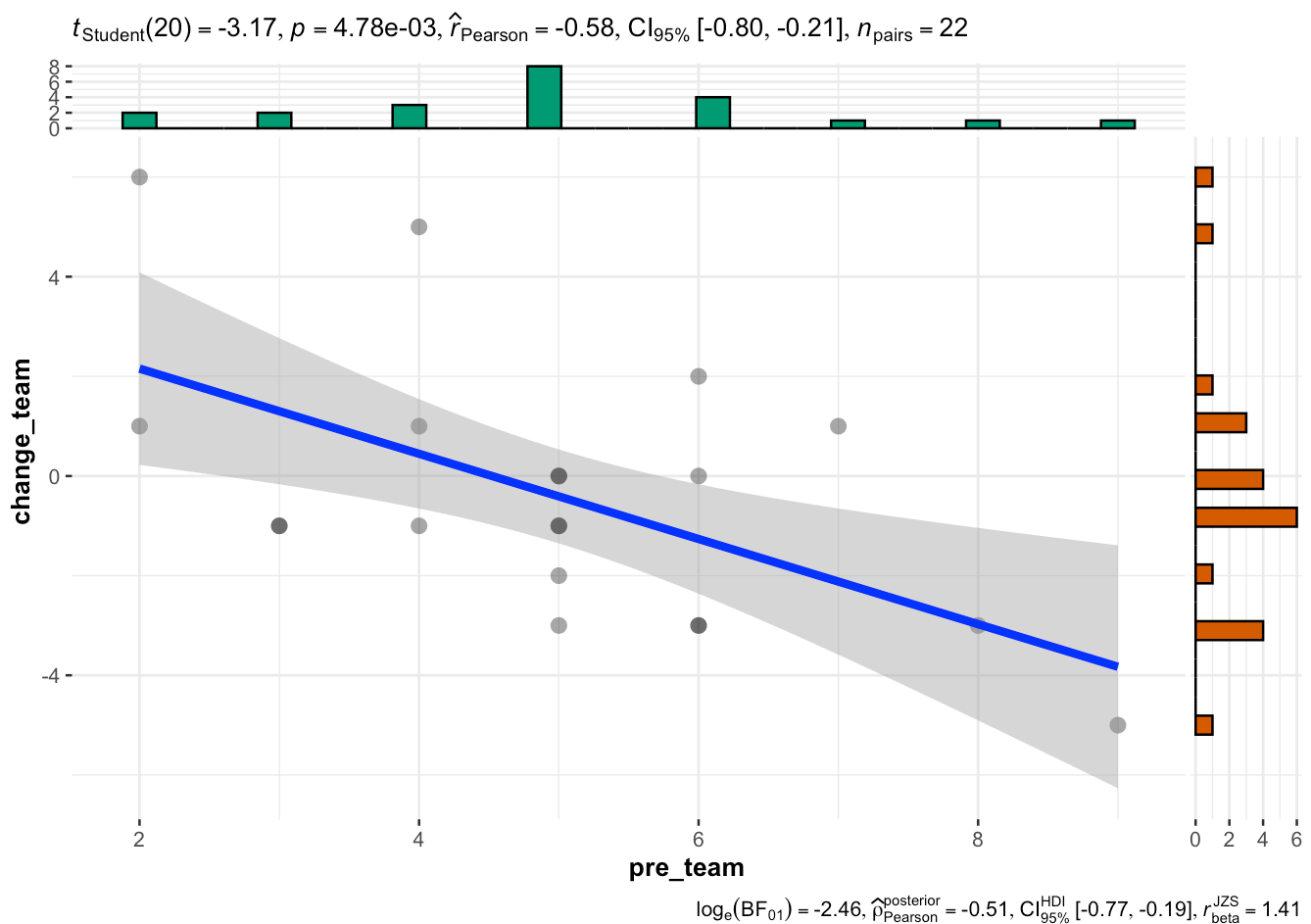
Paired t-test

```
data: survey_responses$post_con_rhythm and survey_responses$pre_con_rhythm
t = 5.3581, df = 21, p-value = 2.586e-05
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 1.001245 2.271482
sample estimates:
mean difference
 1.636364
```

```
?t.test
```

```
ggstatsplot::ggscatterstats(data = survey_responses, x = pre_team, y = change_team)
```

```
`stat_xsidebin()` using `bins = 30`. Pick better value `binwidth`.
`stat_ysidebin()` using `bins = 30`. Pick better value `binwidth`.
```



```
t.test(survey_responses$post_team, survey_responses$pre_team, paired = TRUE)
```

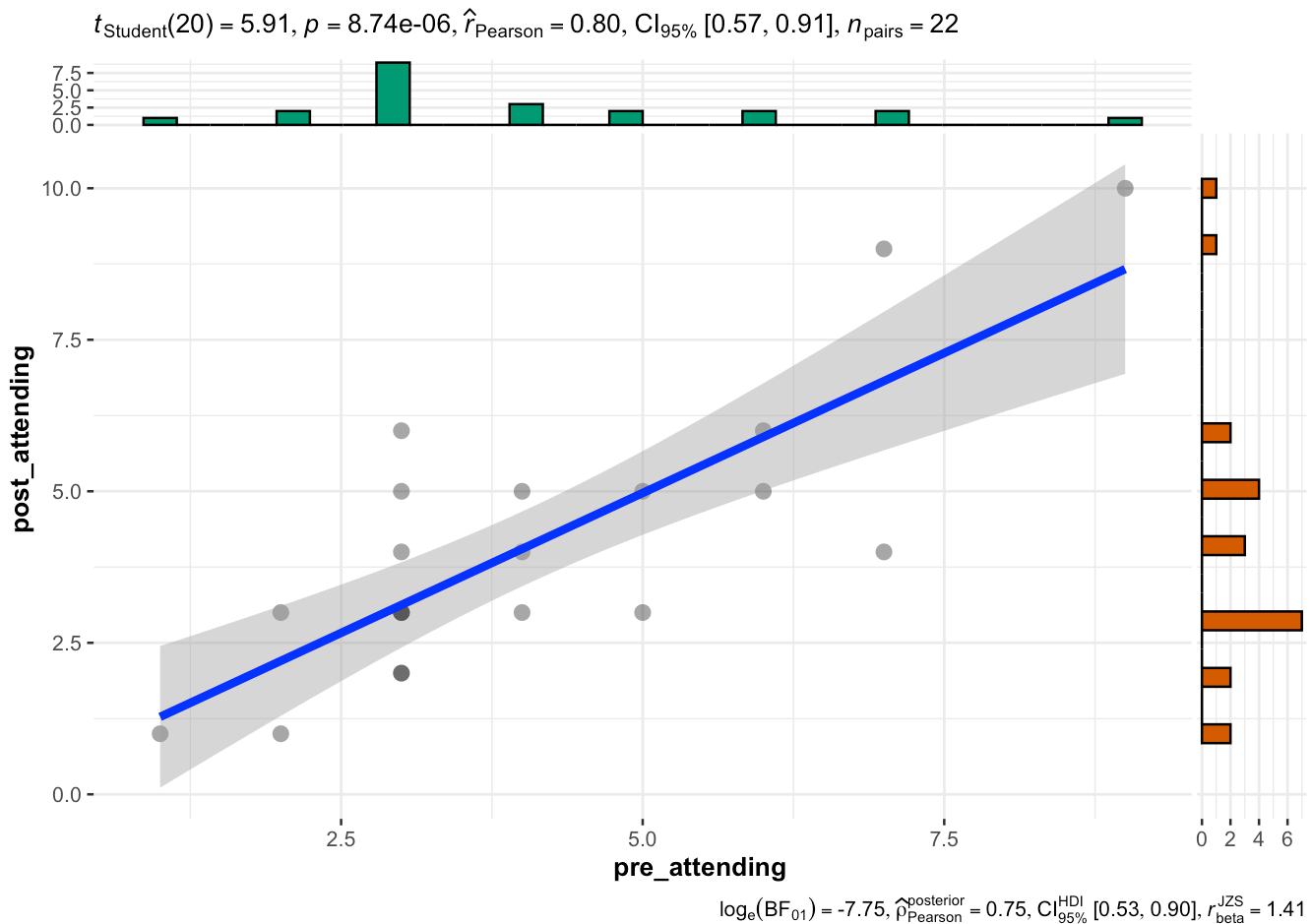

Paired t-test

```
data: survey_responses$post_team and survey_responses$pre_team
t = -0.7559, df = 21, p-value = 0.4581
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -1.5345761  0.7163943
sample estimates:
mean difference
 -0.4090909
```

```
?t.test
```

```
ggstatsplot::ggscatterstats(data = survey_responses, x = pre_attending, y = post_attending,
```

```
`stat_xsidebin()` using `bins = 30`. Pick better value `binwidth`.
`stat_ysidebin()` using `bins = 30`. Pick better value `binwidth`.
```



```
t.test(survey_responses$post_attending, survey_responses$pre_attending, paired = TRUE)
```

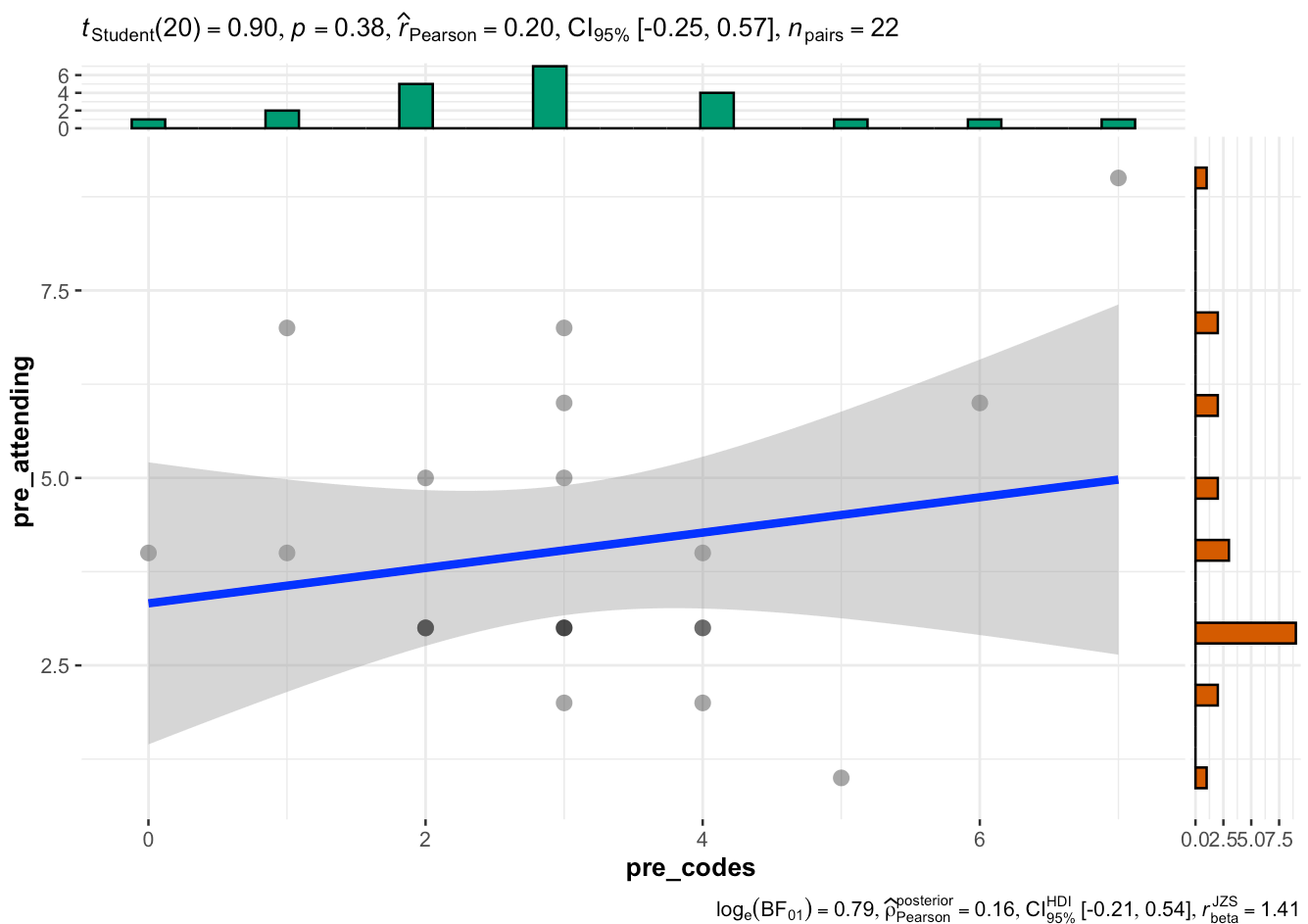
Paired t-test

```
data: survey_responses$post_attending and survey_responses$pre_attending
t = 0.15654, df = 21, p-value = 0.8771
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -0.5584117  0.6493208
sample estimates:
mean difference
 0.04545455
```

```
?t.test
```

```
ggstatsplot::ggscatterstats(data = survey_responses, x = pre_codes, y = pre_attending)
```

```
`stat_xsidebin()` using `bins = 30`. Pick better value `binwidth`.
`stat_ysidebin()` using `bins = 30`. Pick better value `binwidth`.
```



```
t.test(survey_responses$pre_attending, survey_responses$pre_codes, paired = TRUE)
```

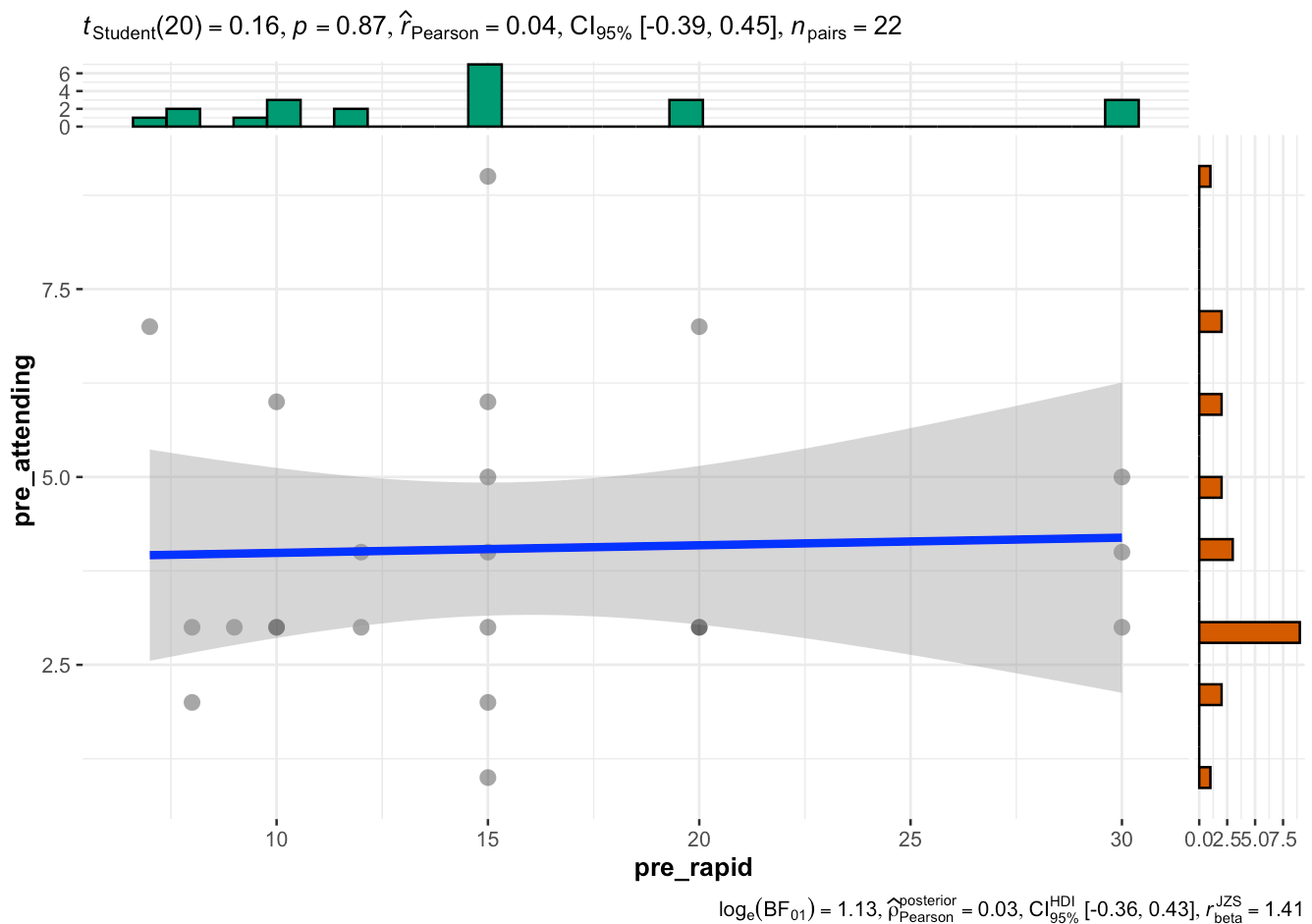
Paired t-test

```
data: survey_responses$pre_attending and survey_responses$pre_codes
t = 2.0683, df = 21, p-value = 0.05115
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -0.005480355  2.005480355
sample estimates:
mean difference
1
```

```
?t.test
```

```
ggstatsplot::ggscatterstats(data = survey_responses, x = pre_rapid, y = pre_attending)
```

```
`stat_xsidebin()` using `bins = 30`. Pick better value `binwidth`.
`stat_ysidebin()` using `bins = 30`. Pick better value `binwidth`.
```



```
t.test(survey_responses$pre_attending, survey_responses$pre_rapid, paired = TRUE)
```

Paired t-test

```
data: survey_responses$pre_attending and survey_responses$pre_rapid
t = -7.4456, df = 21, p-value = 2.554e-07
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -14.653898 -8.255193
sample estimates:
mean difference
 -11.45455
```

```
?t.test
```

```
t.test(survey_responses$pre_attending, survey_responses$pre_codes, paired = TRUE)
```

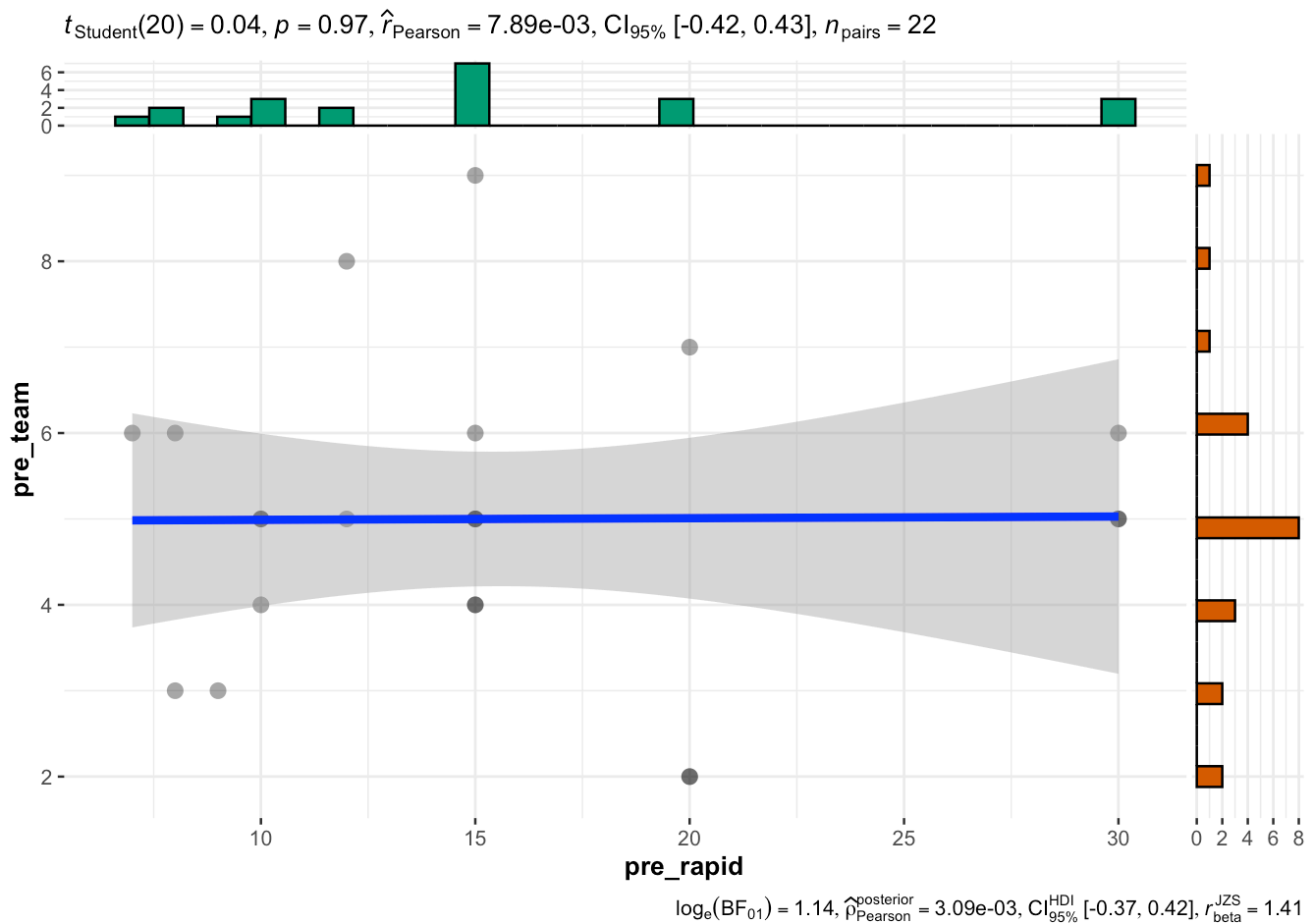
Paired t-test

```
data: survey_responses$pre_attending and survey_responses$pre_codes
t = 2.0683, df = 21, p-value = 0.05115
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -0.005480355 2.005480355
sample estimates:
mean difference
 1
```

```
t.test(survey_responses$pre_team, survey_responses$pre_rapid, paired = TRUE)
```

```
ggstatsplot::ggscatterstats(data = survey_responses, x = pre_rapid, y = pre_team)
```

```
`stat_xsidebin()` using `bins = 30`. Pick better value `binwidth`.
`stat_ysidebin()` using `bins = 30`. Pick better value `binwidth`.
```



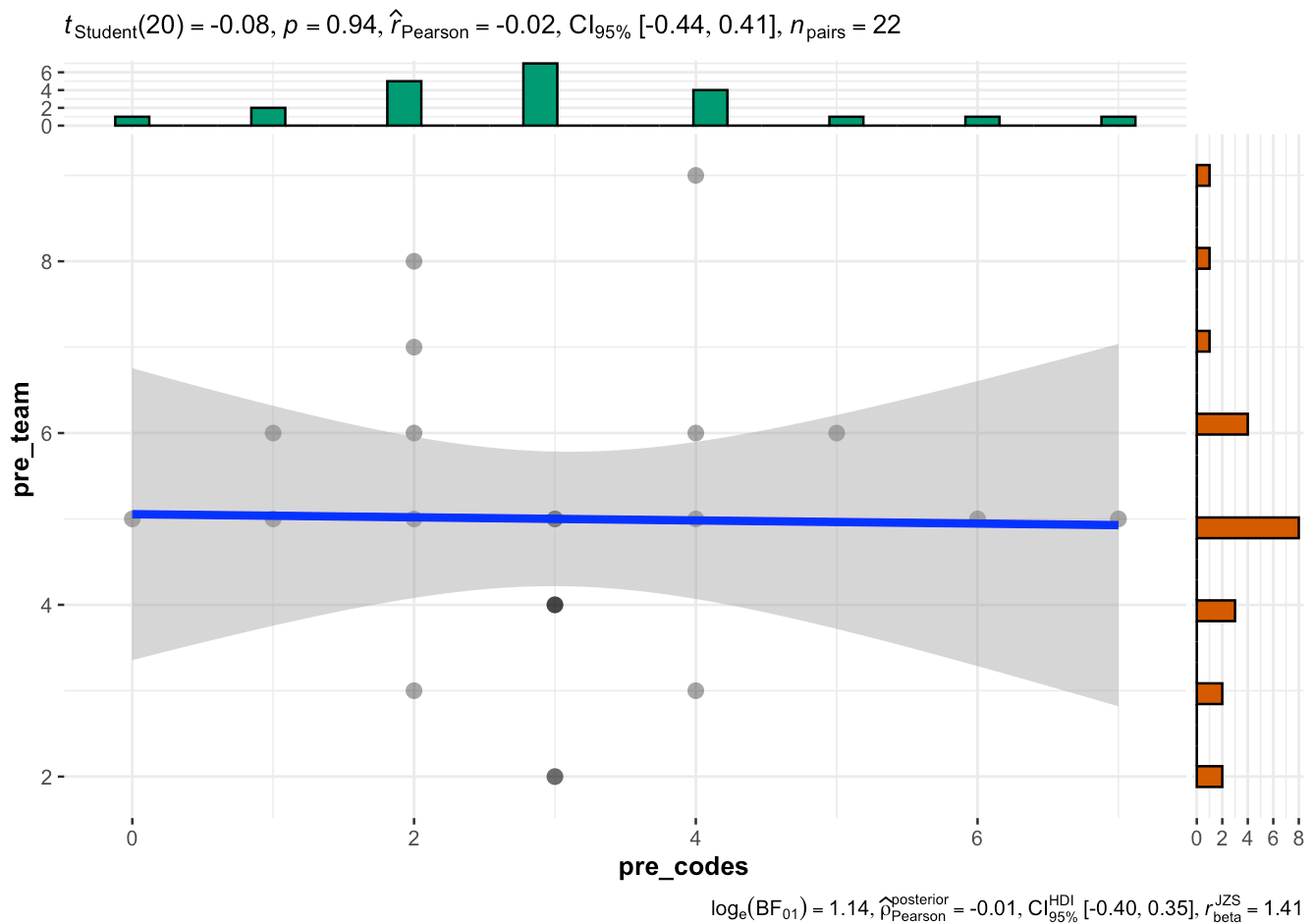
```
t.test(survey_responses$pre_team, survey_responses$pre_rapid, paired = TRUE)
```

Paired t-test

```
data: survey_responses$pre_team and survey_responses$pre_rapid
t = -6.825, df = 21, p-value = 9.517e-07
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -13.699419 -7.300581
sample estimates:
mean difference
 -10.5
```

```
ggstatsplot::ggscatterstats(data = survey_responses, x = pre_codes, y = pre_team)
```

```
`stat_xsidebin()` using `bins = 30`. Pick better value `binwidth`.
`stat_ysidebin()` using `bins = 30`. Pick better value `binwidth`.
```



```
t.test(survey_responses$pre_team, survey_responses$pre_codes, paired = TRUE)
```

Paired t-test

data: survey_responses\$pre_team and survey_responses\$pre_codes

t = 3.8519, df = 21, p-value = 0.0009252

alternative hypothesis: true mean difference is not equal to 0

95 percent confidence interval:

0.8993031 3.0097879

sample estimates:

mean difference

1.954545

```
t.test(survey_responses$pre_team, survey_responses$pre_sim, paired = TRUE)
```

Paired t-test

data: survey_responses\$pre_team and survey_responses\$pre_sim

t = 10.377, df = 21, p-value = 1.009e-09

alternative hypothesis: true mean difference is not equal to 0

95 percent confidence interval:

2.871302 4.310517

sample estimates:

mean difference

3.590909